



David Ricardo & John Maynard Keynes

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David Ricardo

David Ricardo was one of the most influential economists of all time. His main influence was Adam Smith's "The Wealth of Nations". Ricardo became wealthier through stock exchange like his father before he turned 30. Ricardo was also a politician, and a member of the Parliament of Great Britain and Ireland. Ricardo wrote his first economics article at age 37, advocating for a reduction in the note-issuing of the Bank of England. He was also an abolitionist and believed in the autonomy of a central bank as the issuer of money.

Ricardo was known for his meticulous speculation of economic theories. He showed his excellence by constructing abstract model of how capitalism worked and depicted logical implications. His economic theorizing established a style of abstract deductive economic models that has dominated economic theory down to the present

Ricardo was to exert a powerful influence on both the radical Marxist and the conservative neoclassical traditions of economic theorizing throughout the remainder of the nineteenth and twentieth centuries.

Ricardo was a close friend of Malthus. He shared the same view with Malthus about working class nature and causes for poverty and population theory. He was widely influenced by French Revolution, the industrial revolution, the increasing unrest of the working class, and the struggle between the English capitalists and landlords.

Although there were certain primary social issues where Malthus and Ricardo differed with each other. Ricardo was a consistent defender of the interests of the capitalist class. The principal theoretical issues on which they differed were the theory of value and Malthus's theory of gluts.

In his book 'Principles of Political Economy and Taxation', Ricardo stated what he saw as the central problem of political economy:

'The produce of the earth-all that is derived from its surface by the united application of labour, machinery, and capital, is divided among three classes of the community, namely, the proprietor of the land, the owner of the stock of capital necessary for its cultivation, and the labourers by whose industry it is cultivated'. To determine the laws which regulate this distribution is the principal problem in Political Economy.'

In 1815, Malthus published '*An Inquiry into the Nature and Causes of Rent*' where Malthus's theory of rent gave explicit statement to ideas that were already implicit in Ricardo's theory of profit. Malthus complemented a theory of profits on which Ricardo had been working for some time. He had already concluded that the price of corn, relative to the prices of manufactured commodities, was regulated by the tendency for labor and capital, applied to successively less fertile lands, to be less and less productive of corn. He had also concluded that the rate of profit was governed by the diminishing productivity of agricultural labor.

Three weeks after the publication of Malthus's pamphlet, Ricardo published '*An Essay on the Influence of a Low Price of Corn on the Profits of Stock*', showing the Inexpediency of Restrictions on Importation. In it, he first developed the essentials of his theory of distribution.

The Theory of Rent and First Approach to Profits:

In his book, "*An Essay on the Influence of a Low Price of Corn*", Ricardo defined Rent, Net produce, what to consider when determining rent etc.

In Ricardo's opinion, s "that portion of the produce of the earth which is paid to the landlord for the use of the original and indestructible powers of the soil."

- I. His theory of the determination of rent was based on two assumptions: that land differed in its fertility and that all lands could be arrayed along a spectrum from the most fertile to the least fertile.
- II. Competition always equalized the rate of profit among the capitalist farmers who rented land from the landlords.

He also gave statement about Net Produce, "total quantity produced minus all of the necessary costs of production, including the replacement of capital used up in production and the wages of workers." That shows net produce is nothing but the total of surplus value, created by labor, which could go to either profit or rent.

"It is only ... because land is not unlimited in quantity and uniform in quality, and because, in the progress of population, land of an inferior quality ... is called into cultivation, that rent is ever paid for the use of it. When, in the progress of society, land of the second degree of fertility is taken into cultivation, rent immediately commences on that of the first quality, and the amount of that rent will depend on the difference in the quality of these two portions of land. When land of the third quality is taken into cultivation, rent immediately commences on the second,

and it is regulated as before by the difference in their productive powers. At the same time, the rent of the first quality will rise, for that must always be above the rent of the second by the difference between the produce which they yield with a given quantity of capital and labour. With every step in the progress of population, which shall oblige a country to have recourse to land of a worse quality, to enable it to raise its supply of food, rent, on all the more fertile land, will rise. Thus suppose land-No. 1, 2, 3-to yield with an equal employment of capital and labour a net produce of 100, 90, and 80 quarters of corn As soon as population had increased as to make it necessary to cultivate No. 2 ... rent would commence on No. 1; for either there must be two rates of profit on agricultural capital, or ten quarters ... must be withdrawn from the produce of No. 1 for some other. Whether the proprietor of the land, or any other person, cultivated No. 1, these ten quarters would equally constitute rent; for the cultivator of No. 2 would get the same result with his capital whether he cultivated No. 1, paying ten quarters rent, or continued to cultivate No. 2, paying no rent. In the same manner it might be shown that when No. 3 is brought into cultivation, the rent of No. 2 must be ten quarters ... whilst the rent of No. 1 would rise to twenty quarters; for the cultivator of No. 3 would have the same profits whether he paid twenty quarters for the rent of No. 1, ten quarters for the rent of No. 2, or cultivated No. 3 free of all rent”

Ricardo constructed a theory based on the insufficiency of quality land, growing population and the uneven competition among the capitalist farmers who wanted to earn more profit.

He said, as the land is not unlimited and population is also growing, to eradicate the food scarcity, less fertile land must be brought into cultivation. Comparing the two cultivation land, as the second land enter, rent immediately commences on the first land and the amount of that rent will depend on the difference in the quality of these two portions of land. As the third land entered, rent immediately imposed on the previous, second land. The rent of the first quality will rise as well, for that must always be above the rent of the second by the difference between the produce which they yield with a given quantity of capital and labor.

It was competition among capitalist farmers that assured that rent would progress in this way. Suppose there are lands no. 1, 2, 3 which yield with an equal employment of capital and labour a net produce of 100, 90, and 80 quarters of corn. The farmer on number 1, in Ricardo's example, paid only 15 quarters rent after number 3 was brought into cultivation. In that event, he would be making 85 quarters profit (100 quarters net produce minus 15 quarters rent) on the same capital with which the other

two capitalist farmers were making only 80 quarters profit. The other two capitalist farmers could increase their profits by offering to pay landlord number 1 more rent, say 18 quarters, to allow them to farm his land. But as long as the rent on land number 1 was below 20 quarters, capitalists would continue to have an incentive to bid the rent up. Only when the rent was bid up to 20 quarters would they no longer have such an incentive. At that point, the rate of profit would be the same for all capitalist farmers.

Ricardo believed that, in general, competition would tend to equalize the rate of profit for all capitalists. The "restless desire on the part of all the employers of stock," he wrote, "to quit a less profitable for a more advantageous business has a strong tendency to equalize the rate of profits of all." It will give no opportunity to the capitalist farmers to choose more fertile land always to earn more profit. Every cultivated land had to earn same profit and the rent would be imposed on the more fertile lands, less fertile lands could even have all the profit paying zero land.

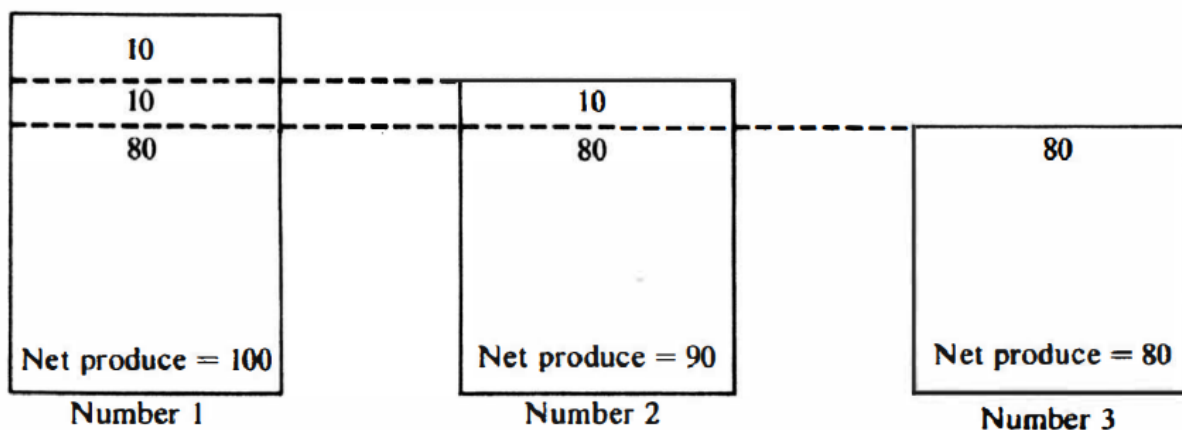


Figure 1: Net Produce and Rent on Three Tracts of Land

The geometrical areas of above three bars represent the net produce in Ricardo's example. Net produce is comprised of profit plus rent.

If only number 1 is farmed, the capitalist farmer makes 100 quarters profit. If number 2 is brought into cultivation, competition bids the rent up to 10 quarters in number 1, and each capitalist earns 90 quarters profit. If number 3 is brought into cultivation, competition bids the rent up to 10 quarters on number 2 and to 20 quarters on number 1, and each capitalist earns 80 quarters profit. As additional land is brought into cultivation, the quantity of land that we label as a parcel is arbitrary.

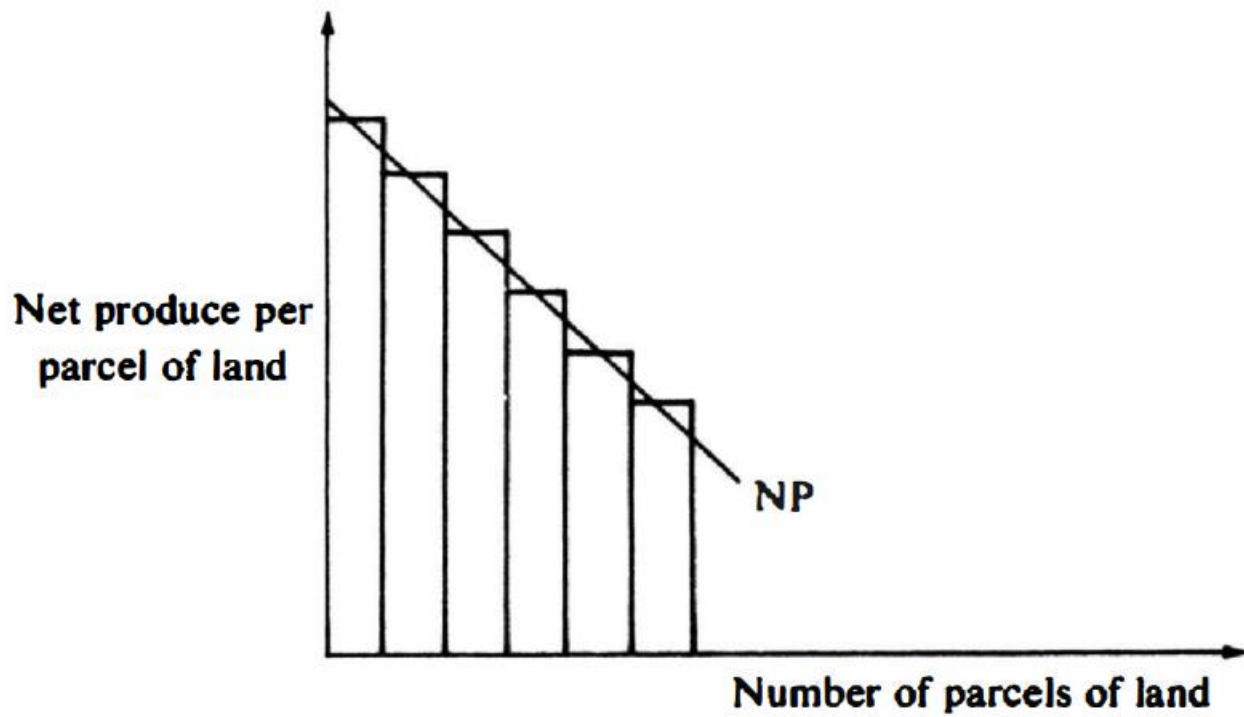


Figure 2: Diminishing Productivity in Agriculture

In this figure above, we have showed net produce per parcel of land on the vertical axis and number of parcels of land on the horizontal axis. The line NP showed only net produce (profit plus rent plus wages).

As we have assumed that land gets continuously less fertile, we could subdivide the land into smaller and smaller parcels, each successive parcel yielding a smaller net produce than the previous parcel. From the figure we can see, as the size of the parcels gets smaller and smaller, the step like tops of the bars come closer and closer to being a simple downward sloping line. We can assume that each land unit is so small that a straight line can be used to show the diminishing fertility of land. NP line shows that net produce per small unit of land diminishes as the quantity of land under cultivation increases. If we assume that wages are the only cost of production, then wages paid per unit of land farmed can be added to line NP in the figure to show total produce.

In the next figure we can see the total produce, line P, showing total produce for any quantity of land as the amount of land in cultivation is increased.

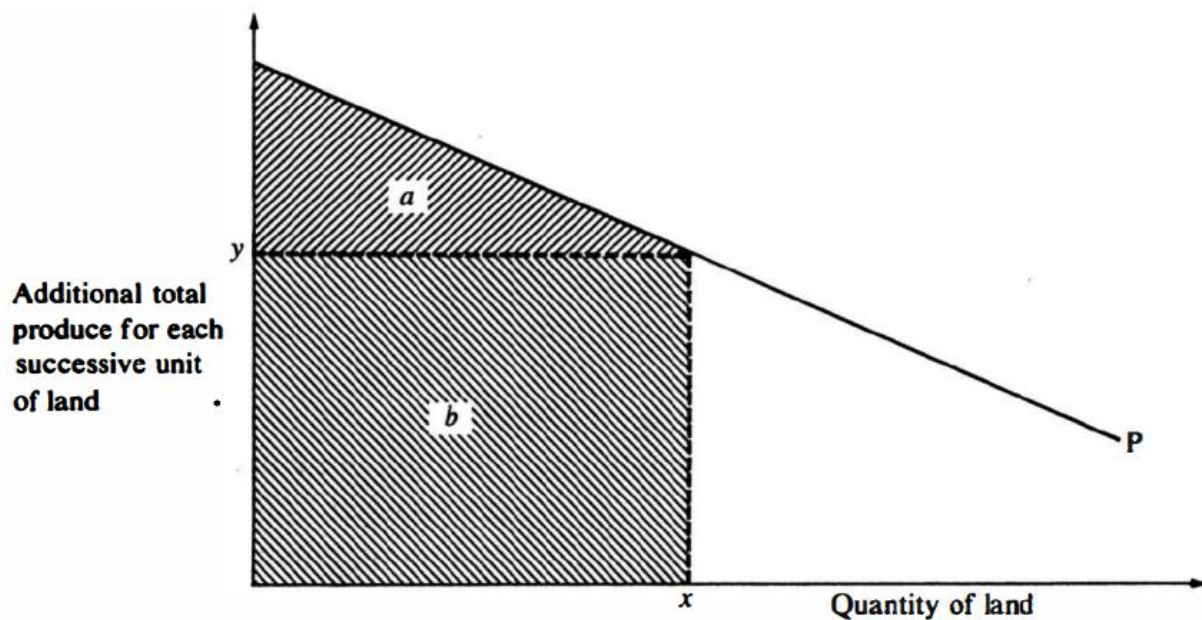


Figure 3: Separation of Rent from Profits and Wages

If x units of land are under cultivation, then y will be the total produce on the last (or no-rent) small unit of land brought into cultivation. The area of triangle a will be the total amount of rent received by the landlord class; the area of rectangle b will be the total profits and wages received by the capitalists and laborers in agriculture.

Ricardo's Theory of Profit:

In his first approach, he assumed a simple economy, consisting of landlords, capitalists, and laborers, which produced only corn. Ricardo saw profits as a surplus. Competition equated the profits of all of the capitalist farmers cultivating the superior grades of land with those accruing to the capitalist farmer cultivating the no-rent, marginal land. Therefore, profits would be determined by the determinants of the profit of the capitalist cultivating the no-rent land.

Ricardo agreed with Malthus that population growth would tend to force the wages of labor to the subsistence level. On this basis Ricardo simplified that, profit was simply what was left over after paying wages. If it's a single commodity model, the capital consisted simply of the corn, which the capitalist "advanced" to the laborers as wages. Accordingly, the rate of profit was the ratio of the net produce on the no-

rent land to wages, both expressed in corn. As long as the net produce was decreasing with each less fertile plot of land brought under cultivation, and as long as the corn-wage rate remained the same, the rate of profit (the net produce in corn over the wage in corn) had to decrease. This statement of profits has been known as, Ricardo's **"Corn Theory of Profit."**

For the manufactured commodities Ricardo expanded the theory, "if population increases decreased the rate of profit in agriculture, and if the rate of profit was determined solely by the productivity of labor and capital on the no-rent land, and if competition equalized all rates of profit, then it followed that the rate of profit in the manufacturing sector as well as in agriculture depended solely on productivity on the no-rent land."

Economic Basis of Conflict between Capitalists and Landlords:

Ricardo's contention in the "Essay" that "the interest of the landlord is always opposed to the interest of every other class in the community." Ricardo identified economic prosperity with capital accumulation and the economic growth and prosperity that this accumulation fostered (as did all of the classical economists). When capitalists earned profits they accumulated capital, which would result in an increased demand for labor. The increased demand for labor led to an increase of the market wage rate above the natural wage rate (subsistence), which led to an increase in population. As long as capitalists continued to make profits, this sequence could repeat itself over and over again. As long as the sequence repeated itself, the economy would be growing, there would be general prosperity, and the wages of labor would be above the subsistence level. But the economy ran into difficulty because of diminishing productivity in agriculture, which caused rents to squeeze out profits.

Ricardo's reasoning is illustrated in the graph below,

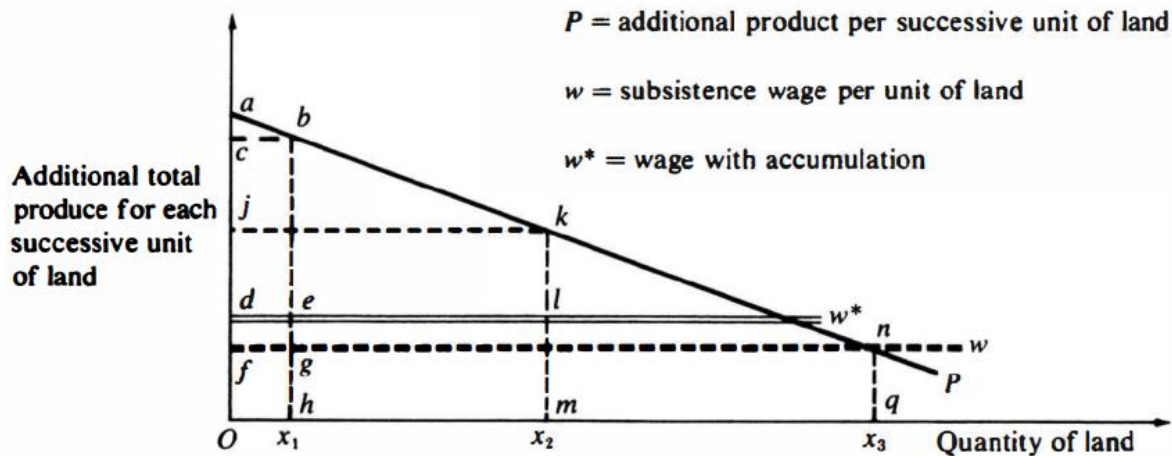


Figure 4: Changes in the Distribution of Income as the Margin of Cultivation Is Extended

In this figure, w has been added to show the subsistence wage that must be paid to the labourers farming one unit and w^* has been added to show the somewhat higher wage that will prevail as long as capital accumulation is occurring. Various points on the graph have been lettered to permit us to illustrate our point. Assume that we observe the economy at a point where x_1 units of land are under cultivation. Assume also that accumulation has taken place in the past and the wage rate is at w^* . Now at x_1 the total quantity of the produce going to rents would be the area of the triangle abc . Wages would be the area of the rectangle $ohed$ (with the rectangle $fged$ representing the excess of wages over the amount necessary for subsistence). Profits would be the residual, or the area of rectangle $debe$. With wages at w^* , above subsistence, population growth will take place. This will require that more land be brought under cultivation. Now suppose that population has grown to the point where x_2 is the amount of land under cultivation. At this point, wages are now given by the area of the rectangle $omld$, rent is the area of triangle akj , and profits are the area of rectangle $dlkj$. Notice that while the wage rate has remained the same, total profits as a share of the total product, as well as the rate of profit, have declined substantially.

It is easy to see in Figure 4 that there is an ultimate limit to this economic growth. Once the economy has brought x_3 amount of land under cultivation, wages will have been pushed back to subsistence (w); rent will be the area of triangle anf , and wages

will be the area of rectangle oqnf. There will be no profits, and, thus, wages will be back at the subsistence level. This explains why, in the struggle of the landlords and capitalists for the surplus or net product, Ricardo believed that diminishing productivity in agriculture would cause profits to be squeezed out steadily by higher and higher rents. Thus, in his "Essay", Ricardo stated that rent was "in all cases a portion of the profits previously obtained on the land. It was never a new creation of revenue, but always a part of a revenue already created.

In Ricardo's model, rent was not directly responsible for squeezing out profit. Rather, it was the increases in the cost of labor created by the increase in the cost of corn, labor's principal staple for subsistence. Ricardo had to show how the increase in wages redistributed a larger and larger share of the net product from profit to rent. To do this he assumed a constant average level of prices (or a constant purchasing power of money). With his belief that competition equalized all rates of profit, then it followed that when the prices of corn and labor increased, prices would have to adjust to equalize the rate of profit in the different sectors of the economy. The labor embodied in producing corn had increased because labor became less productive as the margin of cultivation was extended. This lowered profits in the agricultural sector. But the productivity of labor remained the same in manufacturing, and, therefore, the labor embodied in manufactured goods did not change. In order for competition to equate the rates of profit, it would therefore be necessary for the prices of most manufactured goods to decline relative to corn. With Ricardo's assumption of a constant average level of prices, the increase in the prices of agricultural products would have to be offset by a decrease in the prices of at least some manufactured goods. The effect of these price changes would be the reestablishment of a uniform rate of profit in both sectors but at a lower profit rate. Each increase in the margin of cultivation would thus result in a further decline in the general price level of manufactured goods (all prices, including agricultural prices, again remaining at the same average level) and a decline in the general rate of profit. The decline of profits meant a decline in the rate of accumulation and, hence, a retardation of economic growth and a decrease in general social welfare.

On the basis of these arguments, Ricardo opposed the Corn Laws. By prohibiting the importation of grains, the British government was causing the agricultural sector to bring successively less fertile land under cultivation.

This process was reducing profits and would, if continued long enough, bring economic progress to a halt. In their debate over the Corn Laws, Ricardo's argument

was certainly more coherent and logical than that of Malthus, even though it generally failed to impress most of the members of Parliament who represented the interests of the landlords.

The Labor Theory of Value

Ricardo began his *Principles* by stating that while all commodities having value had to have utility, or they would otherwise not be marketable, utility did not regulate value. He stated: "Possessing utility, commodities derive their exchangeable value from two sources : from their scarcity, and from the quantity of labour required to obtain them." On the next page he stated that scarcity was important only for those commodities that were not freely reproducible.

Ricardo developed the theory by first stating it as the simplified hypothesis that the prices of commodities were strictly proportional to the labor embodied in them in the production process. He then went to some length to describe how this simple principle would have to be modified due to a variety of special circumstances. He believed that these modifications were fully explainable in a systematic and coherent way and, therefore, did not constitute arguments against the labor theory of value, but rather showed the complexity and realism of the theory.

Ricardo began by approvingly quoting Adam Smith's previously cited assertion:

"If among a nation of hunters, for example, it usually costs twice the labour to kill a beaver which it does to kill a deer, one beaver should naturally exchange for or be worth two deer. It is natural that what is usually the produce of two days or two hours labour should be worth double of what is usually the produce of one day's or one hour's labour."

Unlike Smith, he believed that this assertion was as valid for a capitalist society as it had been in the "early and rude" state of society. In capitalist society, however, several qualifications and modifications of the assertion of simple proportionality of labor embodied and prices were necessary. Before making these modifications, Ricardo discussed and then dismissed two objections to the labor theory of value. These were, first, that one could not combine different types of labor having differing skills and differing rates of wages; and second, that the labor theory did not account for the increased productivity made possible by natural resources and capital.

Ricardo's answer to the charge that the labor theory did not consider increases in productivity made possible by land and capital, however, was more adequate and remains to the present time an integral part of the labor theory of value. Tools and machinery, he argued, were intermediate products of labor, created only because

they contributed to the ultimate end of producing a commodity for consumption. Production was a series of labor exertions that effected a transformation of natural resources from the unusable forms in which they existed prior to human activity into forms that had use value. Without an environment to transform, production could not take place; that is, human beings could not even exist. But to consider the environment itself to be productive was to attribute human activity to inert matter. Production, and hence the creation of exchange value, was a strictly human endeavor involving only labor.

Now Ricardo was certainly aware that rent was paid to the owners of natural resources; indeed, as we have seen, a large portion of his *Principles* was devoted to analyzing rent. But rent was strictly a social method of distributing the produce of labor. Production remained solely an activity of human beings. In terms of human costs, he was certainly correct in stating that natural resources perform "their work gratuitously." He quoted, with complete agreement and approval, the following sentences from Adam Smith: "The real price of everything . . . is the toil and trouble of acquiring it . . . Labour was the first price-the original purchase money that was paid for all things. "

Natural resources, then, were the objects that labor transformed in production. But they were simply gratuitously there and were not a social cost of production. Capital was merely so many products of human labor, representing resources that were only partially transformed into their ultimately usable forms. A loom, for example, was produced by labor only to aid in the further production of cloth. Therefore, a loom embodied some of the labor that was ultimately to be embodied in the cloth. In this light, a loom could be seen as merely so much partly produced cloth. To produce was a human activity. Instead of saying, as later neoclassical economists were to say, that the weaver and the loom each contributed to the production of cloth, Ricardo said that the weaver and the laborer who produced the loom each contributed to the production of cloth.

In Ricardo's words:

"In estimating the exchangeable value of stockings, for example, we shall find that their value, comparatively with other things, depends on the total quantity of labour necessary to manufacture them and bring them to market. First, there is the labour necessary to cultivate the land on which the raw cotton is grown; secondly, the labour of conveying the cotton to the country where the stockings are to be manufactured, which includes a portion of the labour bestowed in building the ship in which it is conveyed, and which is charged in the freight of the

goods; thirdly, the labour of the spinner and the weaver; fourthly, a portion of the labour of the engineer, smith, and carpenter, who erected the buildings and machinery, by the help of which they are made; the labour of the retail dealer, and of many others, whom it is unnecessary further to particularize. The aggregate sum of these various kinds of labour determines the quantity of other things for which these stockings will exchange, while the same consideration of the various quantities of labour which have been bestowed on those other things will equally govern the portion of them which will be given for the stockings."

In his recognition of the fact that the contribution of machinery to production was really only the contribution of past labor, Ricardo was repeating Smith's insight, an insight that has always served as the starting point of the labor theory of value.

Adam Smith's Idea on Prices

Adam Smith mentioned that prices should be proportional to the quantities of labor employed for production. He also realized that in order to maintain the relation of prices with labor, it was necessary that the ratio of profits to wages for each commodity must be same. However, competition led the profits on different capitals to be equal. So, equal ratio of profits to wages necessary implied an equal ratio of capital to labor in the production of each commodity. But Smith realized different industry requires different amount of capital per worker and this difference always exists. Eventually Smith had abandoned the idea that the quantity of labor employed would determine the price of a commodity. So, he used a simple cost-of-production theory of prices.

Determinant of Commodity Prices by Ricardo

Ricardo and Marx independently believed that the rate of profits on different capitals would be equal through competition. They also believed that the natural price, or the equilibrium price, was equal to the cost of production. It is to be noted that both capital and labor received the socially average rates of wages and profits. But Ricardo and Marx came to a conclusion that wages and prices are also prices (or derived from other prices). So, there are no other determinant that can explain as the cause of price. Ricardo's explanation confirmed by the labor theory of value which asserted that the value of a commodity will be determined by the labor employed in the specific commodity.

Ricardo showed the labor theory of value is consistent even with different ratios of capital to labor and changes in the wage rate. In order to prove the point, he explained how the labor theory could be modified.

Effect on Price due to Changes in Wage Rate

Let us imagine two capitalist firms. One of the firms consists of a fund intended to be used only for wages of the labor. On the other hand, the other firm consists of expensive machines and a small portion of fund to pay the wages. We assume the rate of profit in the first period of production is 10 percent. If each firm employs 100 workers, price of the commodity of firm 1, say P_1 , is

$$P_1 = (\text{Wages of the 100 workers}) + (10 \text{ percent profit on wage fund})$$

,and price of the commodity of firm 2, say P_2 , is

$$P_2 = (\text{Wages of the 100 workers}) + (10 \text{ percent profit on wage fund}) + (\text{Cost of the machines used in production}) + (10 \text{ percent profit on the cost of the machines})$$

It is assumed that $P_2 = 2P_1$. Now, if the wage rate increases in the next production period, with the same level of output and same amount of labor using the same technologies, profits will decline. According to the labor theory of value, even with higher wages, as the amount of labor is unchanged, the relative prices should stay the same. However, the relative price of the commodities also changes.

For the first commodity's cost the share of wages was 90 percent, and the share of profits was 10 percent. The increase in wages will have a very large effect on the new price of the first commodity, and the decrease in the rate of profit will have a relatively small effect. The first commodity's price will undoubtedly go up substantially. On the one hand, wages constituted a relatively small percent of the second commodity's cost, so the impact of the increased wages on its total costs will be relatively small. The costs of the machinery it uses may go up or down, depending on the impact of the wage increase on the firm that produces these machines. But included in the costs of the second commodity are the profits on both the wages fund and the expensive machinery used in its production. Therefore, the effect of the new lower rate of profits will be much greater on the costs of the second commodity than on those of the first commodity. Therefore, as we have assumed $P_2 = 2P_1$, will no longer hold. So, it can be said that beyond doubt that price ratio has changed while the quantities of labor used in the production of both products remain unchanged.

Ricardo explained this situation by stating three situations using the labor theory. In these three situations, prices would not be proportional to labor embodied.

- i) Different proportions of capital (that is, machineries, tools and buildings) used in different productions.

- ii) The durability of the machines and tools used in production.
- iii) The duration of the capital tied up in the production process

Different Ratios of Capital-to-Labor Ratio

The above discussion leads us to the same conclusion that the proportion and varying duration of the capital used will alter the prices of commodities even with the same amount of labor used in production. So, we can characterize the situations in either two ways. They are

- i) Different ratio of capital to labor used in production
- ii) Time dimension

Let us now illustrate the effect of differing ratios of capital to labor.

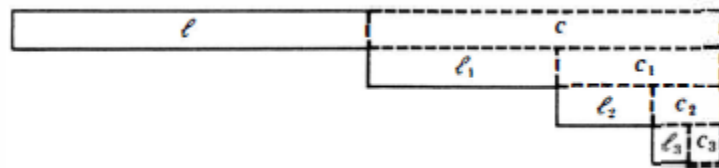


Figure 5: Reduction of Capital to Past Labor

Capitals, that is, machineries, tools etc. used in production process are also some produced commodities from earlier period. So, we can reduce the capitals into past labor. The above figure represents how capital can be reduced to a series of time-dated labor inputs. The bars on the first row represents the inputs used in the present period, labor (l) and capital (c). The commodities were produced in the previous period by labor (l_1) and commodities (c_1). These commodities were in turn produced in the period before that by labor (l_2), and commodities (c_2), and so on.

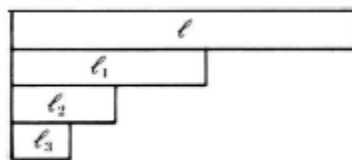


Figure 6: Sequence of Time-Dated Labor

Let us consider the commodities used in the production, that is, capitals, are sufficiently small in magnitude. So, we find only a series of time-dated labor

depicted in the above figure. In the figure, the present labor is labeled (l); past labor has a subscript, indicating how many periods into the past the labor was exerted.

If the wage costs of (l_3) are \$100 and the rate of profit is 10 percent throughout the period, then the capitalists have two choices. They can invest the \$ 1 00 at the beginning of a series of one-year production processes and then reinvest each year the entire amount, including all profits earned in the previous period ; or they can invest in the four-year project illustrated in the figure.

If the capitalist chooses the first alternative, he or she will receive at the end of the first year = $(100 + 100 \times 0.10) = \110 . Reinvesting the entire amount, he or she will receive = $(110 + 110 \times 0.10) = \121 at the end of the second year and $(121 + 121 \times 0.10) = \133.10 at the end of the third. Therefore, if the four-year project is to be equally profitable, the commodities representing that part of capital embodying the labor (l_3) must be valued at \$133.10 at the beginning of the final year, and the profits in the final year's production must be = $(133.10 \times 0.10) = \$13.31$ on this portion of capital. So the labor (l_3), which originally costs the capitalist \$100, finally results in \$146.41 ($133.10 + 13.31$) being added to the price of the commodity. Of this, \$100 repays the capitalist the initial \$100 wage expense, and \$46.41 is considered profit for having \$100 in capital tied up for four years. The capitalist ends up with the same amount if he or she invests in four separate one-year ventures with a return of 10 percent profit, provided that each year the capitalist reinvests the original \$100 plus all profits earned to that point. This is why, in Ricardo's terms, either more capital per worker or a longer period of production would give the same result. If the capitalist had \$100 tied up for only two years , the value of that capital at the beginning of the final year would be \$110 (\$100 plus \$10 profit from the preceding year); whereas if the capitalist had \$100 tied up for four years , the value of that capital at the beginning of the final year would be \$133.10 (\$100 plus \$10 profit the first year, plus \$11 profit the second year, plus \$12.10 profit the third year). Thus, the value of capital is greater in the four-year process than in the two year process, even though the initial wage costs (and hence the original labor embodied in the capital) were the same in each case.

Numerical Example

We have seen how differences in capital to labor ratio make difference in commodity prices and how wage rate alter such prices. Now, let us look at a numerical illustration of this situation. It will make a clear understanding of the actual theory.

Let us consider two firms, Firm 1 and Firm 2. Firm 1 produces commodity X and Firm 2 produces commodity Y. The production techniques of both the firms are illustrated by the following figure.



a) Labor units necessary to produce 100 units of commodity X. (400 units of labor over a period of five years)

b) Labor units necessary to produce 100 units of commodity Y. (400 units of labor over a period of three years)

Figure 7: Two Production Processes Having the Same Quantity of Labor but Differing Compositions of Capital

In part (a), 400 units of embodied labor produce 100 units of commodity X, and in part (b), 400 units of embodied labor produce 100 units of commodity Y. It requires 100 units of present labor (labeled l) and 300 units of past labor (labeled c), applied uniformly at the rate of 75 units per year over the four previous years, to produce X. It requires 300 units of present labor and 100 units of past labor, applied at the rate of 50 units per year over the two previous years, to produce Y. Obviously, if prices were proportional to labor embodied, the two prices would be equal.

i) When wage rate is \$1.00 and profit rate is 50 percent:

For Commodity X----

Labor Cost = (Wages) $\times$ (number of labor units) = $(100 \times \$1) = \100

Cost of Machinery (cost of past labor compounded each year by the rate of profit)

For 75 $l_4 = (\$75 + 75 \times 0.50) = \112.5

For 75 $l_3 = (\$112.5 + \$75) + (\$112.5 + \$75) \times 0.50 = \$281.25$

For 75 $l_2 = (\$281.25 + \$75) + (\$281.25 + \$75) \times 0.50 = \$534.375$

For 75 $l_1 = (\$534.375 + \$75) + (\$534.375 + \$75) \times 0.50 = \$914.06$

So, Cost of Machinery = \$914.06

Profit Cost (rate of profit times sum of labor cost and machinery cost) = $0.50 \times (100 + 914.06) = 507.03$

Total Cost = $\$(100+507.3+914.06) = \1521.36

Price per unit = $\$ (1521.36/100) = 15.21$

For Commodity Y----

Labor Cost = (Wages) $\times$ (number of labor units) = $(300 \times \$1) = \300

Cost of Machinery (cost of past labor compounded each year by the rate of profit)

For 50 $l_2 = \$50 + 50 \times 0.50 = \75

For 50 $l_1 = (\$75 + \$50) + (\$75 + \$50) \times 0.50 = \$187.50$

So, Cost of Machinery = \$187.50

Profit Cost (rate of profit times sum of labor cost and machinery cost) = $0.50 \times (300 + 187.5) = 243.75$

Total Cost = $\$(300+187.50+243.75) = \731.25

Price per unit = $\$ (731.25/100) = 7.31$

ii) When wage rate is \$2.00 and Profit rate is 10 percent:

For Commodity X----

$$\text{Labor Cost} = (\text{Wages}) \times (\text{number of labor units}) = (100 \times \$2) = \$200$$

$$\text{Cost of Machinery (cost of past labor compounded each year by the rate of profit)} \\ = \$765.78$$

$$\text{Profit Cost (rate of profit times sum of labor cost and machinery cost)} = 0.10 \times \\ (200 + 765.78) = 96.58$$

$$\text{Total Cost} = \$(200+765.78+96.58) = \$1062.36$$

$$\text{Price per unit} = \$ (1062.36/100) = 10.62$$

For Commodity Y----

$$\text{Labor Cost} = (\text{Wages}) \times (\text{number of labor units}) = (300 \times \$2) = \$600$$

$$\text{Cost of Machinery (cost of past labor compounded each year by the rate of profit)} \\ = \$231.00$$

$$\text{Profit Cost (rate of profit times sum of labor cost and machinery cost)} = 0.10 \times \\ (600 + 231.00) = \$83.10$$

$$\text{Total Cost} = \$(600+231.00+83.10) = \$914.10$$

$$\text{Price per unit} = \$ (914.10/100) = 9.14$$

Three important observations can be made in comparing the results above. First, changing the wage rate substantially alters the relative prices of X and Y; whereas the price of X is more than double that of Y. This illustrates the fact that it is the difference in profits that causes prices not to be proportional to labor embodied. Second, if the wage rate increases, the consequent decrease in the profit rate will lower the prices of those commodities in which the profit component of costs is large. The third important point is this: as long as the rate of profit is positive, commodity X will always have a higher price than commodity Y. Commodity X also has a higher ratio of capital to labor than commodity Y.

Distribution of Income and Labor theory of value

The distribution of income refers to the way in which the total income of a society is distributed among individuals or groups. One of the main factors that determine the distribution of income is the Labor Theory of Value. In this essay, we will explore the concept of distribution of income and how it is related to the Labor Theory of Value.

Distribution of Income

The distribution of income is determined by various factors such as education, skills, experience, gender, and race. These factors affect an individual's ability to earn income. For example, individuals with higher levels of education and skills tend to earn more income than those with lower levels of education and skills. Similarly, experience plays a role in determining income, as individuals with more experience in a particular field tend to earn more income than those with less experience. However, despite these individual factors, the overall distribution of income in a society is heavily influenced by the Labor Theory of Value.

Labor Theory of Value

The Labor Theory of Value is a concept that suggests that the value of a good or service is determined by the amount of labor that goes into producing it. According to this theory, the value of a good or service is not determined by its utility or demand but by the amount of labor that is required to produce it. This theory was first developed by classical economists such as Adam Smith, David Ricardo, and Karl Marx.

The Labor Theory of Value suggests that individuals who contribute more labor to the production of goods and services should receive a greater share of the income generated by those goods and services. For example, in a factory setting, the workers who contribute more labor to the production of a product should receive a greater share of the income generated by that product than those who contribute less labor.

According to Ricardo, the value of a good or service was determined by the amount of labor that was required to produce it. He believed that the value of a good or service was not determined by the amount of money that was paid for it, but by the amount of labor that was required to produce it.

Ricardo argued that the distribution of income in society was determined by the relative scarcity of different factors of production, particularly land and labor. He

believed that if land was in short supply relative to labor, then landowners would be able to command a higher share of the total income in society. On the other hand, if labor was in short supply relative to land, then workers would be able to command a higher share of the total income in society.

Ricardo's theory of the distribution of income was based on the idea that the value of a good or service was determined by the amount of labor that was required to produce it. He believed that the relative scarcity of different factors of production, particularly land and labor, would determine the distribution of income in society.

In Ricardo's view, the profits earned by capitalists were also determined by the relative scarcity of land and labor. If there was a shortage of labor relative to land, then capitalists would have to pay higher wages to attract workers, which would reduce their profits. On the other hand, if there was a shortage of land relative to labor, then capitalists would be able to charge higher prices for their products, which would increase their profits.

Ricardo's labor theory of value was based on the idea that the value of a good or service was determined by the amount of labor that was required to produce it. He believed that this theory was more objective and scientific than the subjective theory of value, which held that the value of a good or service was determined by the utility that it provided to the consumer.

Implications of the Labor Theory of Value

The Labor Theory of Value has significant implications for the distribution of income in a society. According to this theory, individuals who contribute more labor to the production of goods and services should receive a greater share of the income generated by those goods and services. For example, consider two workers in a factory who produce the same product but have different levels of skill and experience. According to the Labor Theory of Value, the worker who contributes more labor to the production of the product, either through working longer hours or using more advanced skills, should receive a greater share of the income generated by that product. Therefore, in a fair and just distribution of income, the worker who contributes more labor should earn a higher wage than the worker who contributes less labor.

However, in reality, the distribution of income is often skewed, with some individuals and groups receiving a disproportionate share of the income generated

by the economy. For example, in a capitalist economy, the owners of capital, such as large corporations and wealthy individuals, often have more power and influence over the distribution of income than workers who contribute labor to the production of goods and services. This power and influence can result in the owners of capital receiving a greater share of the income generated by the economy, even if they contribute less labor than workers.

The Labor Theory of Value suggests that individuals who contribute more labor to the production of goods and services should receive a greater share of the income generated by those goods and services. However, in reality, the actual distribution of income is often skewed, with some individuals and groups receiving a disproportionate share of the income generated by the economy. **This discrepancy between theory and reality can be attributed to various factors such as power and wealth disparities, institutional biases, and market forces.**

Power and Wealth Disparities

In a capitalist economy, the owners of capital, such as large corporations and wealthy individuals, often have more power and influence over the distribution of income than workers who contribute labor to the production of goods and services. This power and influence can result in the owners of capital receiving a greater share of the income generated by the economy, even if they contribute less labor than workers.

For example, consider a large corporation that produces a product. The workers who contribute labor to the production of the product may receive a small share of the income generated by that product, while the owners of the corporation receive a large share of the income. This is because the owners of the corporation have more power and influence over the distribution of income than the workers who contribute labor.

Institutional Biases

Institutional biases can also affect the distribution of income in a society. For example, gender and race can be institutional biases that affect the distribution of income. Women and people of color may receive lower wages than white men, even if they contribute the same amount of labor to the production of goods and services. This is because of institutional biases that favor white men in the labor market.

Market Forces

Market forces can also affect the distribution of income in a society. For example, in a market economy, the demand for certain goods and services can affect the wages that workers receive. If there is high demand for a certain type of labor, such as software engineering, then workers in that field may receive higher wages than workers in fields with lower demand, even if they contribute the same amount of labor.

However, market forces can also be influenced by power and wealth disparities. For example, large corporations may use their power and influence to suppress wages in certain industries, even if the demand for labor in those industries is high.

Solutions for Ensuring a Fair Distribution of Income

While there are various factors that can affect the distribution of income, there are also solutions that can be implemented to ensure a fairer distribution of income. Some of these solutions include:

1. **Progressive taxation:** Progressive taxation is a tax system where individuals with higher incomes pay a higher percentage of their income in taxes than individuals with lower incomes. This can help to reduce the wealth disparity and ensure a fairer distribution of income.
2. **Labor unions:** Labor unions are organizations that represent workers in negotiations with employers. By bargaining collectively, labor unions can help to ensure that workers receive fair wages and benefits.
3. **Minimum wage laws:** Minimum wage laws are laws that set a minimum wage that employers must pay to their workers. This can help to ensure that workers receive a minimum level of income.
4. **Universal basic income:** Universal basic income is a system where all citizens of a country receive a basic income from the government. This can help to reduce poverty and ensure that all individuals have a basic level of income.
5. **Education and training:** Education and training can help individuals to acquire skills that are in demand in the labor market. This can increase their earning potential and ensure that they receive a fair share of the income generated by the economy.

The distribution of income in a society is an important issue that has significant implications for economic and social outcomes. The Labor Theory of Value suggests that individuals who contribute more labor to the production of goods and services should receive a greater share of the income generated by those goods and services. However, in reality, the distribution of income is often skewed, with some individuals and groups receiving a disproportionate share of the income generated by the economy. This can be attributed to various factors such as power and wealth disparities, institutional biases, and market forces. To ensure a fairer distribution of income, solutions such as progressive taxation, labor unions, minimum wage laws, universal basic income, and education and training can be implemented. By implementing these solutions, we can work towards creating a more just and equitable society where all individuals have the opportunity to share in the benefits of economic growth.

The Impossibility of Gluts

Malthus' idea that insufficient aggregate demand caused cyclical gluts or depressions was the foundation for his recommendation that landlords be given a bigger part of national production. Accordingly, this theory was challenged by Ricardo. He reasoned that subsidizing the landlord's useless expenditure would be as profitable to the capitalist as having a fire in the capitalist's warehouse destroy some of his merchandise. Second, Ricardo contended that the forces of supply and demand would automatically adjust prices and the composition of aggregate output, preventing a widespread glut.

The reasoning is straightforward. It states that capitalists produce what they do not need in order to swap it for something they do. Money facilitates the exchange, but money is not wanted in and of itself. A producer creates a commodity in order to trade it for another commodity. When a commodity is exchanged for money, the producer wants to exchange the money for another commodity. No one wants to hoard money because it has no beneficial properties other than the ability to acquire another commodity. As a result, production generates its own demand. A capitalist has a one-dollar desire for other goods for every dollar in commodities produced. Ricardo presented his reasoning succinctly:

No one creates unless he intends to consume or sell it, and no one sells unless he intends to buy another commodity that will be immediately helpful to him or will contribute to future production. By producing, one becomes either the purchaser and

consumer of his own commodities or the purchaser and consumer of the goods of another person.

Commodities are always purchased with commodities or services; money is simply the instrument of transaction. Too much of a given item may be produced, resulting in a market glut that does not repay the capital invested in it; however, this cannot be said of all commodities.

Say's Law

Say's law states that the production of goods creates its own demand.

In 1803, John Baptiste Say explained his theory.

“It is worthwhile to remark that a product is no sooner created than it, from that instant, affords a market for other products to the full extent of its own value.”

This view suggests that the key to economic growth is not increasing demand, but increasing production. Say's views were expanded on by classical economists, such as James Mill and David Ricardo.

Why does supply create its own demand?

If a businessman produces a good, then he will be keen to sell it. This production creates wages for workers and income for the businessman. Therefore, the production has increased wealth and leads to demand for other goods.

Say argued it was irrational to hoard money because ‘he is most anxious to sell it immediately, lest its value should diminish in his hands’ i.e. inflation may reduce the value of cash.

This theory assumes that markets clear and that businessmen produce goods that are demanded by the market.

Machinery as a cause of involuntary unemployment:

Unemployment can be voluntary or involuntary. Involuntary unemployment is when a person wants to work at the going wage rate, but can't find available jobs as there is no demand for their labor we can say it is a situation where workers are willing to work at the market wage or just below but are prevented by factors beyond their

control. Example of involuntary unemployment include: Layoffs: When a company needs to reduce its workforce, it may lay off some of its employees. These individuals may be involuntarily unemployed until they are able to find new employment. Ricardo then stated that he still believed capitalists and landlords would benefit by such a change in productive technology, but "that the substitution of machinery for human labor is often very injurious to the interests of the class of laborers." This was because laborers initially would be diverted from the production of wage goods to the production of capital goods. In the following period, there would be fewer wage goods and hence a reduced demand for labor because the demand for labor was limited by the availability of wage goods.

When the new machinery was put into use, it would require some laborers, but not nearly as many would be put back to work as were previously fired, because the machinery was constructed only in order to reduce the number of laborers necessary to produce a given output and hence to reduce the capitalist's wage costs and increase profits. Thus, the net income of society (profits and rents) could be increased while the gross income (profits, rents, and wages) was being reduced. In that case, many laborers would "be thrown out of employment," and a large part of the working class would "become redundant compared with the funds which are to employ it. Ricardo concluded that the opinion entertained by the laboring class, that the employment of machinery is frequently detrimental to their interests, is not founded on prejudice and error, but is conformable to the correct principles of political economy."⁵⁷ Such a conclusion meant that he agreed with Malthus that the market might not be very effective in reallocating resources when a change in the conditions of production took place, and that the result could be a chronic depression in the labor market that would reduce the total produce of the economy. It also meant that Ricardo's faith in the accumulation of capital as the principal force that would increase the economic welfare of all society was unfounded. But his main sympathy was clearly with the capitalist class, not with society as a whole. He concluded the chapter by asserting: The statements which I have made will not, I hope, lead to the inference that machinery should not be encouraged. "This hope was not based on a plan to ameliorate the conditions of workers. For he had stated, and he still believed, that like all other contracts, wages should be left to the fair and free competition of the market, and should never be controlled by the interference of the legislature." Ricardo had to assume that if England imported the relatively more expensive commodity, then the higher price of that commodity would be a reasonable index of the increase in the "sum of enjoyments" in order for his conclusion to follow from

his premises. To put it another way, suppose that landlords and capitalists were the only people who drank wine, and that laborers did not have enough clothing to keep them warm. From a labor theory point of view, the circumstances that led to the expropriation of a large part.

The Ricardian Theory of Comparative Advantage and International Trade:

Ricardo's widely known and impactful comparative advantage theory suggests that nations can gain an international trade advantage when they focus on producing goods that produce the lowest opportunity costs as compared to other nations.

This theory basically suggests, a country should increase production and export of that good in which country has lower opportunity cost. and a country's ability to produce goods at lower opportunity cost is called comparative advantage.

Ricardo was the first economist to argue consistently that free international trade could benefit two countries, even though one country produced all of the traded commodities more efficiently than the other. He was also one of the first economists to argue that, because capital was relatively immobile between nations, a separate theory of international trade was needed. Ricardo argued that a country need not have an absolute advantage in the production of any commodity in order for international trade between it and another country to be mutually beneficial. By absolute advantage was meant greater efficiency in production, or the use of less labor in production. Two countries could both benefit from trade if each had a relative advantage in production. By relative advantage was meant simply that the ratio of the labor embodied in two commodities differed between two countries, such that each country would have at least one commodity where the relative amount of labor embodied would be less than that of the other country. Following table is a reproduction of Ricardo's example to illustrate the principle of comparative advantage. In this table, Portugal has an absolute advantage in the production of both wine and cloth; that is, fewer labor hours are required to produce either commodity in Portugal than are required in England. If we assume that the prices of wine and cloth are proportional to the labor embodied in them in both England and Portugal, then the ratios of the two prices in each country will be the same as the ratio of labor hours required to produce the commodities in each country.

In Portugal it takes ninety hours to produce one unit of cloth and eighty hours to produce one unit of wine. This means that wine requires only 88 percent as much labor as cloth and that the price of wine is only 88 percent that of cloth. In England, the labor embodied in and the price of wine are both 120 percent that of cloth. Thus,

Portugal uses relatively less labor to produce wine, and the price is relatively lower. On the other hand, Portugal uses 112 percent as much labor to produce cloth as wine, and the price of cloth in that country is 112 percent that of wine. England uses only 83 percent as much labor to produce cloth as wine, and, again, the price of cloth is only 83 percent that of wine. Thus, England uses relatively less labor to produce cloth, even though it uses absolutely more labor; therefore, England has a relative advantage in producing cloth.

Now suppose that the Portuguese have been producing only wine (the commodity in which they have a comparative advantage) and decide that they want cloth as well. They have two ways of getting the cloth diverting some labor from the production of wine to the production of cloth or trading wine with England for cloth. Suppose that if trade takes place it does so at the price ratio prevailing in England. To produce one unit of cloth will require ninety working hours. This means that the Portuguese must discontinue producing 1.12 units of wine for every unit of cloth they produce. But if they trade with England.

	Cloth	Wine	Ratio of price of wine to price of cloth	Ratio of price of cloth to price of wine
England	100	120	1.20	0.83
Portugal	90	80	0.88	1.12

Table 1: Number of Hours Required to Produce One Unit of Cloth and Wine in England and Portugal

At the English price ratio, they need only give up 0.83 units of wine for each unit of cloth. Obviously, trading would leave them a greater combined total of wine and cloth than would the production of both commodities. Similarly, if England has been producing only cloth but could trade at the price ratio prevailing in Portugal, then it should not produce wine. To do so would require that it give up 1.2 units of cloth for each unit of wine produced while it could give up only 0.88 units of cloth for one unit of wine if it traded with Portugal. It is obvious that both countries could benefit if each could trade at the price ratio prevailing in the other country. But both countries could benefit if they trade at a price ratio that is between the ratios prevailing in each country. The ratio of the price of wine to the price of cloth is 1.2 in England and 0.88 in Portugal. If both countries trade on a one-to-one basis, one unit of cloth for one unit

of wine, both could consume a larger combined total of wine and cloth than if they each produce both commodities. This, then, explains Ricardo's theory of comparative advantage. On this basis, he argued that free trade would be beneficial to both countries. Every extension of trade would "very powerfully contribute to increase the mass of commodities and the sum of enjoyments."⁶⁰ Every restriction on trade would likewise reduce the "sum of enjoyments." This principle, then, was another link in Ricardo's general attack on the Corn Laws.

So, the Ricardian model of comparative advantage theory shows the possibility that an industry in a developed country could compete against an industry in a developing country.

Social harmony and class conflict:

For Ricardo, class conflict is the inevitable result of a population increase leading to the bringing of marginal lands under cultivation, which leads to wages dropping to a subsistence level. For Marx, on the other hand, class conflict was not a result of population growth but of changing modes of economic production.

In order for Ricardo's conclusion to follow from his premises, he had to assume that if England imported the relatively more expensive commodity, then the higher price of that commodity would be a reasonable index of the increase in the "sum of enjoyments." In other words, suppose wine were drunk only by landlords and capitalists and that laborers did not have enough clothing to keep themselves warm. The labor theory perspective would tend to focus on the circumstances that resulted in so much of the produce of labor being expropriated as rent and profit. The utility theory, however, generally assumes the laws of property ownership and the distribution of wealth to be fixed, or "natural," and, consequently, would tend to focus on the fact that capitalists and landlords would prefer having their surplus in both cloth and wine rather than in cloth alone. Free trade would increase the enjoyments of those who had the wealth and purchasing power but not necessarily the enjoyments of the working class.

Ricardo could have suggested, for example, that the importation of wine be prohibited until every worker had adequate clothing. Some would certainly argue that this would increase social welfare more than importing more wine for the capitalists and the landlords would. But he did not do so, instead insisting that free trade would maximize the "sum of enjoyments".

Most of Ricardo's *Principles* is based on the labor theory, or production vantage point.

Adam Smith's theory of history stopped with capitalism, which he considered the highest possible stage of social evolution. For that reason, Smith tended to take the property relationships and the distribution of wealth prevailing in capitalism as given and fixed. Only when these class differences were taken as given and fixed could Smith argue for the beneficence of the invisible hand. Ricardo's approach to understanding capitalism was much more ahistorical than Smith's. He considered the property relationships, distribution of wealth and power, and class relationships of capitalism to be eternal, unchanging, and natural. The labor theory of value sees society from a perspective that focuses on conflict, as Ricardo's *Principles* so clearly illustrates. But conflict implies change. And change implies that socioeconomic systems, such as feudalism or capitalism, have a life process of their own: they are born; they grow to maturity; and they decay and die. Ricardo's advocacy of the interests of the capitalists had been best served by taking a labor theory perspective. It was therefore inevitable that his writings should involve contradictions on the issue of whether capitalism creates conflict or a harmony of interests. We have already encountered an example where the invisible hand did not work: "The dealings between the landlord and the public are not like dealings in trade, whereby both the seller and buyer may equally be said to gain, but the loss is wholly on one side, and the gain wholly on the other."⁶⁵ Similarly, in discussing international trade, as soon as he did not take the balance of real, coercive power to be given, it became obvious to Ricardo that conflict, not harmony, was the result.

Every argument that Ricardo gave showing how coercive restrictions in the free market could benefit the mother country at the expense of the colony could just as easily demonstrate how exchange between capitalists and laborers could be regulated by coercive restrictions that benefited the capitalist at the expense of the laborer (or vice versa, as many of the leaders in the rising labor movement of Ricardo's time hoped). Given Adam Smith's assertion that civil government, so far as it is instituted for the security of property, is in reality instituted for the defence of the rich against the poor, or of those who have some property against those who have none at all, it behooved both Smith and Ricardo to show why government would not be used by capitalists in exactly this way. Without such a demonstration, the invisible-hand argument would inevitably be used simply to justify any observed outcome in a market, capitalist system.

So, from the discussion of class conflict theory it is learned that David Ricardo argued that attempts to increase or improve workers' wages were pointless because wages would, in time, return to or hover around the subsistence .

John Maynard Keynes

Before understanding Keynesian economics, one should follow the economic ideas before it so that the rationale of Keynesian theory may be clear to him. With that regard first we start with the Utilitarian ideological elements that dominated the previous time.

The three principal ideological elements of neoclassical utilitarianism were:

- (1) The marginal productivity theory of distribution, which pictured competitive capitalism as an ideal of distributive justice;
- (2) The "invisible hand" argument, which pictured capitalism as an ideal of rationality and efficiency;
- (3) The faith in the automatic, self-adjusting nature of the market, which demonstrated that the principal functions of government should be to enforce contracts and to defend the powers and privileges of private property.

All of this are the ideological defense of laissez-faire capitalism. Though the first two arguments are seeming all good for the capitalist economy, the third one may not all the time.

Before capitalist era merchants and landlords used to have abundant financial power, after that they participated in government and for due reasons unliked by capitalist. Much of their unlikeness came from the early capitalist's involvement in corrupt, and monarchy, for example the actions of several European kings, as well as from the actions of the English Parliament. Therefore, under the banner of laissez faire, capitalists could campaign for governments that more effectively promoted unrestricted profit making and yet make these campaigns appear to be humanitarian efforts to promote the general public's freedom from tyrannical government abuse. The argument for self-adjusting markets (Say's law) was an effective argument for limiting the functions of existing governments.

But the capitalist market system has never adjusted smoothly and automatically to full-employment equilibrium as per the theory. We can easily get that picture from history. During the late nineteenth century, the development of worldwide capital markets and improvements in production and transportation led to immense opportunity to gain profit and so giant corporations, trusts, and cartels arose. There were two important consequences of this increased industrial concentration:

- (1) Due to unregulated competition giant corporations' started to struggle
- (2) The anarchy of the market became more severe due to less production.

As a result, depressions became longer and more severe and occurred somewhat more frequently. If we compare two timeline for example during the late eighteenth and early nineteenth centuries, when the capitalists had not completely brought the existing governments under their control, and the late nineteenth and early twentieth centuries, when the governments of the capitalist countries were quite firmly in the controlled by the capitalists, we can surely conclude that capitalist should turn to the government as the only possible means of escaping the anarchy's and depressions of the late nineteenth century. But ironically the capitalist hold positions in the governments and thus influences the regulations and acts, and facilitate their own interest from this sector. As a result, situation worsen for labor organizations and unfair competition persists.

Interestingly from the neo classical point of view these events can be justified also. They can support their argument by saying that the giant corporation creates workable competition, which is good for the economy. Government should play his role in regulation area due to create harmony that is promised by the invisible hand.

But in reality, capitalism situation declines. During the first half of the nineteenth century, for example, the United States had only two severe economic crises (beginning in 1819 and 1837), and England had four (beginning in 1815, 1825, 1836, and 1847). During the last half of the century, the crises became more severe and increased in number to five in the United States (beginning in 1854, 1857, 1873, 1884, and 1893) and six in England (beginning in 1857, 1866, 1873, 1882, 1890, and 1900). In the twentieth century, the situation grew worse. Increasingly frequent depressions halt capitalism, and finally occurs the Great Depression of the 1930s.

The Great Depression of the 1930s was a worldwide phenomenon, affecting every major capitalist economy. In the United States, for example, on October 24, 1929 (a day that came to be known as "Black Thursday"), the New York Stock Exchange saw security values, a phenomenon that eventually destroyed all confidence in business. As a result, businessmen cut back production and investment. This decreased national income and employment, which in turn worsened business confidence even more. Before the process came to an end, thousands of corporations had gone bankrupt, millions were unemployed, and one of the worst national catastrophes in history was under way.

Between 1929 and 1932, there were over 85,000 business failures; more than 5,000 banks suspended operations; stock values on the New York Stock Exchange fell from \$87 billion to \$19 billion; unemployment rose to 12 million, with nearly one-fourth of the population having no means of sustaining themselves; farm income fell by more than half; and manufacturing output decreased by almost 50 percent.

City mayors reported that they may have resources only to go for a while. People tries to get money from whomever they could, ranging from insurance company to relatives. Those who never knew starvation, have to gone into the process deeply.

Natural resources were still as plentiful as ever. The nation still had as many factories, tools, and machines. The people had the same skills and wanted to put them to work. Yet millions of workers and their families begged, borrowed, stole, and lined up for a pittance from charity, while thousands of factories stood idle or operated at far below capacity. The explanation lay within the institutions of the capitalist market system. Factories could have been opened and men put to work, but they were not because it was not profitable for businessmen to do this. In a capitalist economy, production decisions are based primarily on profits not on people's needs.

In this disastrous situation, it became clear to many neoclassical economists (but by no means to all of them) that the myth of the self-adjusting market had outlived its ideological usefulness. The unregulated anarchy of the market was becoming a threat to the very existence of capitalism. It was obvious to many economists that drastic measures were needed on a scale that could be undertaken only by the government. But neoclassical economics, with its emphasis on the automaticity of the market, offered no cure for the problem of capitalism. In neoclassical theory depressions did not occur, so there was no need to remedy them.

That is the time when one of the most brilliant conservative economists of this century: John Maynard Keynes (1883-1946) came in. In his *The General Theory of Employment, Interest and Money*, published in 1936, Keynes attempted to show what had happened to capitalism so that measures could be taken to preserve the system. In his understanding of depressions, Keynes could have examined the writings of either Malthus or Marx. Keynes knew very well which side of the class struggle Marx was on and which side Malthus was on. Marx's ideas, Keynes concluded as a dull one that unfortunately influence people and thus history. But Malthus had also constructed a theory of capitalist depressions, and that was somewhat okay with Keynes.

Theoretical Setting of Keynes's Analysis

The conceptual context of Keynesian theory was alike Walrasian general equilibrium theory. It is an analysis of a continuous process of production, circulation, and consumption.

In a given production period, a firm produces goods. From the proceeds of the sale of these goods the firm pays its costs of production, which include wages, salaries, rent, supplies and raw materials, and interest on borrowed funds. What remains after these costs are paid is profit. Here the cost of production to the business firm represents income to an individual or another firm. The profit is also income-the income going to the owners of the firm. It follows that the value of what has been produced must be equal to the incomes generated in producing it. In terms of the entire economy, the aggregate picture is the same as that for the individual firm: the value of everything produced in the economy during any period is equal to the total of all incomes received in that period. Therefore, in order for businesses to sell all that they have produced, people must spend in the aggregate all their incomes. If an amount equal to the total income in society is spent on goods and services, then the value of production is realized in sales. In that case, profits remain high, and businessmen are willing to produce the same amount or more in the succeeding period. This process can be seen as a circular flow: Money flows from businesses to the public in the form of wages, salaries, rents, interest, and profits; this money then flows back to the businesses when the public buys goods and services from them. As long as businesses sell all that they have produced and make satisfactory profits, the process continues.

But this does not happen automatically. When money flows from businesses to the public, some of it does not flow directly back to the businesses. The circular flow has leakages. To begin with, all people do not spend all of their incomes. A percentage is saved, usually in banks, and therefore withdrawn from the spending stream. There are also two other leakages: people buy goods and services from foreign businesses, so that the money spent on these imports is not spent on domestically produced goods, and the taxes that people pay are withdrawn from the income-expenditure flow. These three leakages (saving, imports, and taxes) may be offset by three spending injections into the income-expenditure flow. First, imports

can be offset by exports. They are exactly offset, for example, when foreigners buy goods produced in the United States in amounts equal in value to imports purchased by Americans. Second, the government uses taxes to finance the purchase of goods and services. If it uses all taxes for this purpose and balances the budget, then government expenditures will exactly offset taxes in the spending stream. Third, if businessmen wish to expand their capital, they can finance investment in capital goods by borrowing the funds that have been saved. Investment, then, may exactly offset the saving leakage. If these three injections into the income-expenditure flow are just as large as the three leakages, then spending equals the value of production.

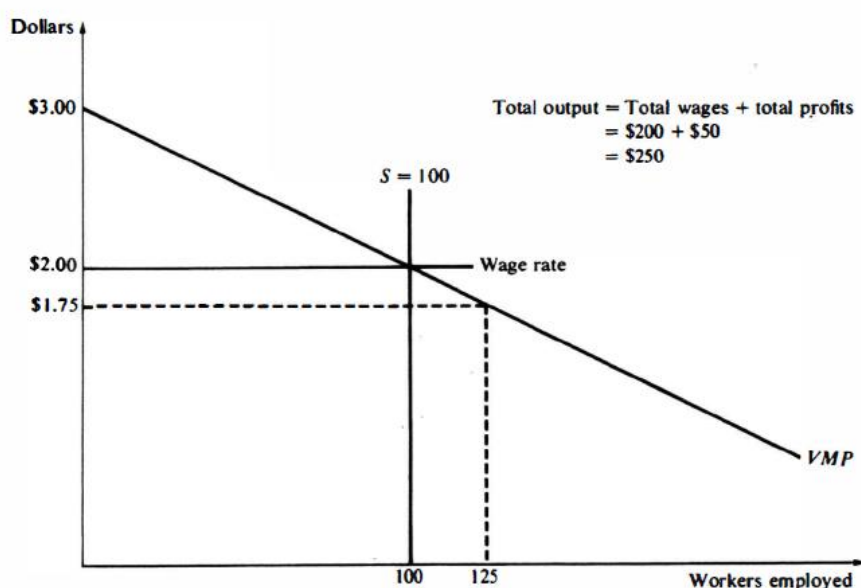


Figure 1: Neoclassical Determination of the Wage Rate and Total Output

This was the orthodox neoclassical vision of how competitive capitalism normally functioned. The levels of total employment and total output were determined by the production function and the free choices of owners of the factors of production. In a short-run period, with a given amount of capital in existence, the demand for labor was determined, by the value of the marginal product of labor. With this given demand for labor, neoclassical theory explained both the wage rate and the amount of total output by the supply of labor. This analysis is illustrated in Figure 1, where VMP is the value of the marginal product of labor and S is the supply of labor. In Figure 1, if 100 workers

wish to work, then they must accept a wage (determined by the value of their marginal product) of \$2.00 per worker. In this case, the total of wages is \$200.00, and \$50.00 goes to profits. Total output in this example has a value of \$250.00. Now if 125 workers wish to work, then the value of the marginal product of labor will decline to \$1.75. For more workers to find employment, the labor rate must decline. If 125 workers are employed, then the total wages are \$218.75 ($125 \times \1.75), the total profits are \$78.12, and the total output is \$296.87.

If unemployment existed in this analysis, it was because workers refused to work unless they received more than the value of their marginal product. For example, in Figure 15.1, if the wage rate were \$2.00, then only 100 workers would be employed. If 125 workers wished to work, then they would have to accept a wage rate of \$1.75. If they were willing to do this, then profit maximizing behavior of capitalists would assure that 125 workers would be employed. If the workers refused to take a wage cut and 25 remained unemployed, then neoclassical economists defined these workers as voluntarily unemployed and insisted that full employment prevailed. Only when workers were willing to work at a wage equal to the value of their marginal product and could not find work at that wage rate were neoclassical economists willing to admit that involuntary unemployment existed. But capitalists maximized profits when they hired workers up to the point where the value of their marginal product equaled the wage rate. Therefore, profit maximization, the neoclassicists concluded, insured that there would never be any involuntary unemployment. When unemployment appeared to exist, it was only because workers refused to take the wage cuts necessary to bring the wage rate into equality with the lower value of the marginal product that would result when more workers were employed.

If workers were willing to accept a wage that was equal to the value of their marginal product, then there would never normally be any problem of aggregate demand being equal to aggregate supply. All income would normally be spent. The three injections into the income-expenditure flow would normally equal the three leakages. This was because: (1) neoclassical theory showed that the free play of supply and demand would balance international transactions; (2) "sound fiscal policy," in the view of both neoclassical economists and most politicians, dictated that the government maintain a balanced budget in which taxes

equaled government expenditures; and (3) the rate of interest would always bring saving and investment into equality.

The last point represented a crucial difference between the neoclassical economists and Keynes, so we will briefly elaborate the neoclassical view.

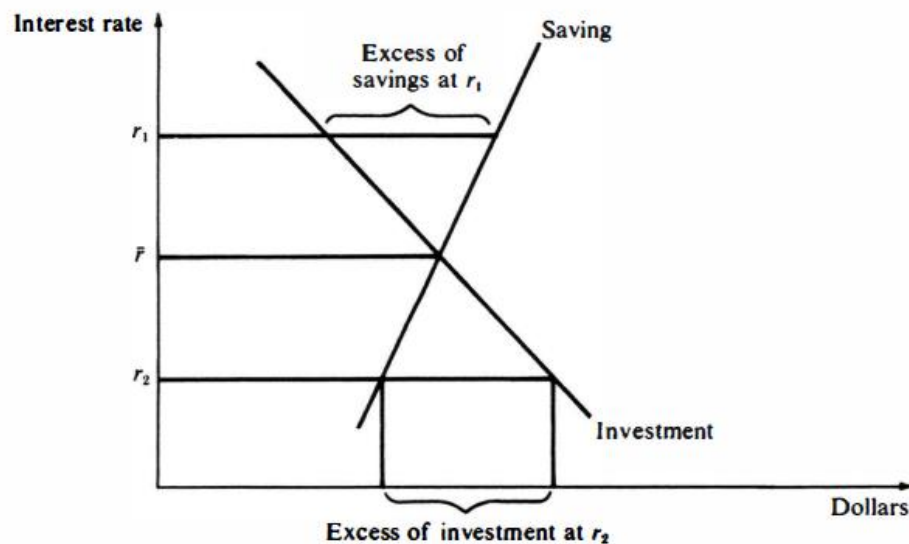


Figure 2: Interest, Saving and Investment

In this view, people always preferred present consumption to future consumption. Therefore, people would only save when they received a reward for doing so, the reward being interest received on their saving. The higher the rate of interest, the more they would save; and the lower the rate of interest, the less they would save. People who invested in a new plant and equipment or other capital goods had to pay for the funds they invested. The amount they had to pay was determined by the rate of interest. The lower the rate of interest, the cheaper would be the funds for investment and more investment would be undertaken. Figure 2 illustrates these two relationships between the rate of interest and saving and the rate of interest and investment. In figure 2, if r_1 is the prevailing rate of interest, then saving exceeds investment (and, aggregate supply exceeds aggregate demand); savers cannot find sufficient investment to absorb all of the funds that they have saved. They will bid competitively to find borrowers for their funds and bid the interest rate down to $\bar{r}$. With the rate of interest falling to equate saving with investment, the aggregate excess supply of commodities will be eliminated. If r_2 is the prevailing rate of interest, then investors cannot find sufficient saving to finance

their investments. They will competitively bid for the available funds and bid the rate of interest up to r . At r saving equals investment.

Therefore, in neoclassical theory, competition automatically created an interest rate at which saving and investment were equal. This meant that all three leakages from the income-expenditure flow would automatically be brought into equality with all three injections into the flow, and aggregate demand into equality with all three injections into the flow, aggregate demand would automatically equal to aggregate supply. The only conceivable cause of what appeared to be involuntary unemployment was, to nearly all neoclassical economists, the refusal of laborers to take sufficiently large cuts in wages.

Keynes's Analysis of Capitalist Depressions

Keynes's theoretical work **General Theory, 1936** was a phenomena after the Great Depression in 1929 which was the result of Capitalism. Keynes addressed capitalized depression through analysis. He assumed in depression, the rate of utilization of the productive capacity of physical capital declined clearly along with wiping out number of employed workers. Keynes stick to the marginal productivity that states wages were always equal to the workers' marginal productivity which wasn't quite right at the time. This is where Keynes against his rest theory because capitalist depressions decreased employment not increasing real wages of workers.

Keynes's departure from Neoclassical Theory:

Keynes suggested that capitalist government would save capitalism with his theoretical insights. It required some neoclassical tenets to be repressed. He was adamant with both marginal productivity theory of distribution and the idea of free market efficiently allocated resources even though both ideologies were knot with the belief that an automated full employment would be created by free market. Keynes dropped the idea which states if a capital economy started from a situation of full employment, then the rate of interest would automatically equate saving and investment and keep aggregate demand equal to aggregate supply.

He rejected the tenet that comprised the neoclassical theory of automaticity were twice:

First, he agreed on the neoclassical idea that saving was influenced by the rate of interest, he emphasized that the level of aggregate income was a far more influence on the amount of saving and investment than was the rate of interest. **Second**, he debated that saving and investment did not determine the rate of interest.

It was a price that equalized the demand and supply of money. Keynes wanted to gain the first result.

He argued that the level of consumption and the level of saving were mainly a function of the level of income. The consumption function shows the relation of saving and consumption to the level of income. The marginal propensity to consume and the marginal propensity to save were both assumed to be less than one and these were not predetermined by the determinants of interest rate.

His second departure from neoclassical theory of automaticity of the market that he rejected the neoclassical theory of determining the interest rate. The rate of interest was determined by the demand and supply of the money. The supply of money was constant due to central monetary policy but not the demand for money. Demand for money determined by three motives- transactions, precautionary and speculative. Speculative motive was related to the rate of interest.

According to Keynes a portion of the demand for money relied on the interest rate in future. When the interest rate rose, money demanded for speculative purposes declines and increased as the interest rate fell.

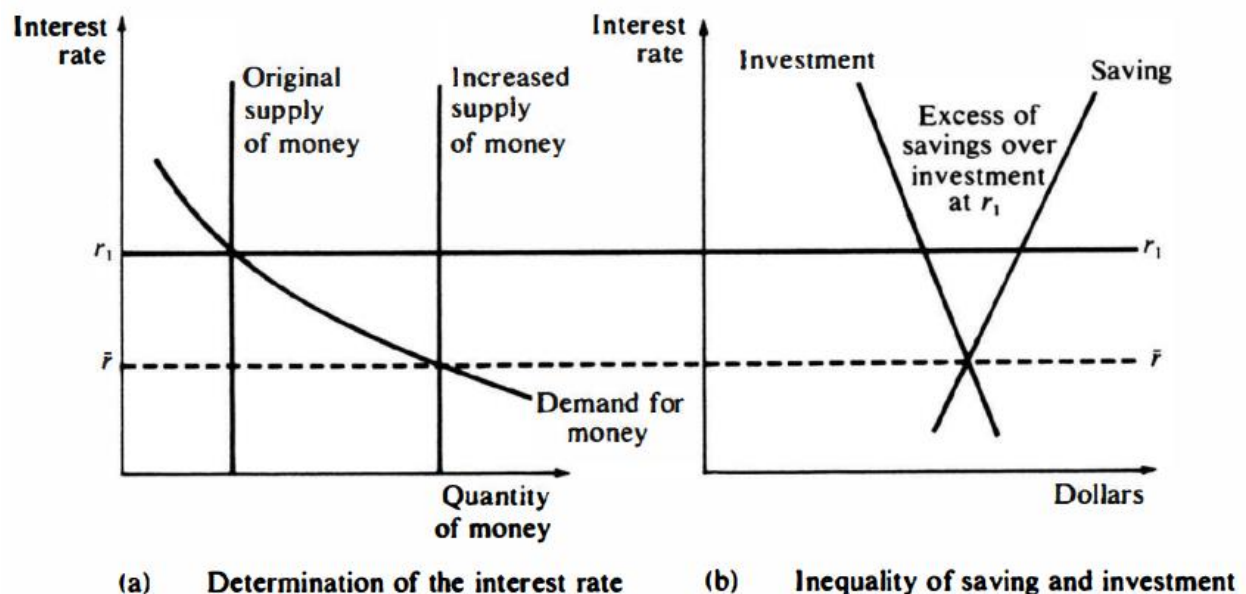


Figure 3: Determination of the Rate of Interest and the Inequality of Saving and Investment

The monetary authorities could increase the supply of money so at that interest rate, saving equaled investment, aggregate demand equaled aggregate supply.

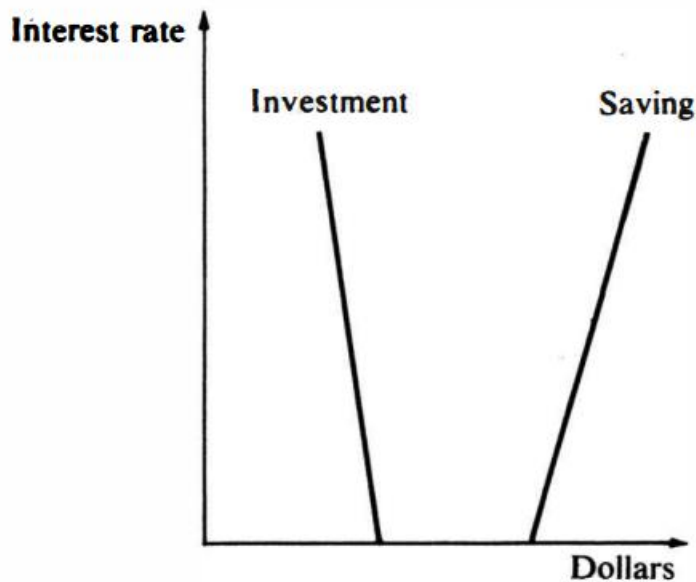


Figure 4: Situation in Which Rate of Interest Cannot Equate Full Employment Levels of Saving and Investment

When the distribution of income is unequal and the full employment level of output and income was very high that no matter how low the interest rate sank, saving and investment could not be equated.

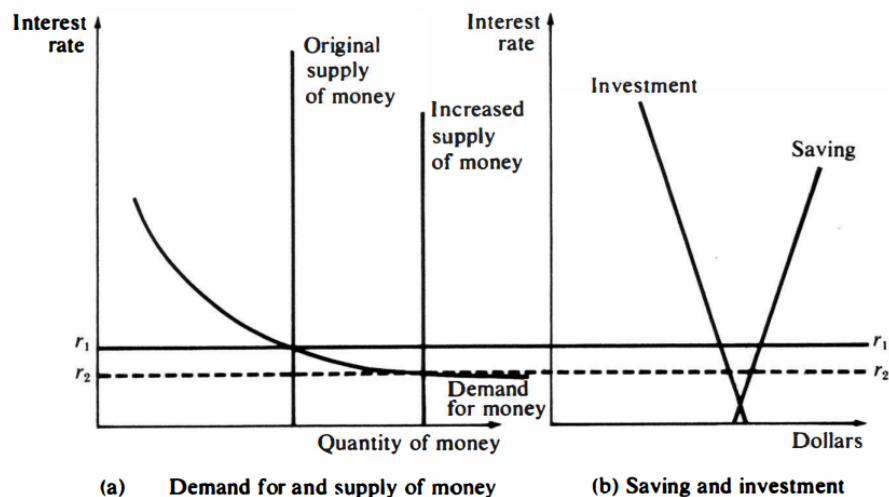


Figure 5: Situation in Which Monetary Policy Cannot Equate Full Employment Levels of Saving and Investment

Keynes argued that it was possible if the interest rate would equate the full employment levels of saving and investment was very low, monetary policy might not be able to lower the interest rate sufficiently. If monetary authorities pushed the rate of interest low so that almost everyone would prefer to hold cash rather than securities even when the monetary authorities increased the amount of money in the system.

Keynes's Defense of the Marginal Productivity Theory of Distribution

Marginal productivity theory aims to define how much should a worker be paid according to their capacity to produce. That is in equilibrium each productive agent will get the reward according to their marginal productivity.

Keynes did not agree with the belief that aggregate demand would always equal to aggregate supply at the full-employment level of income. In 'General Theory' he state that the neoclassical 'theory of employment' was based on two fundamental postulates the first being that the 'wage is equal to the marginal product of labor'. He agreed with this. According to Keynes

'If employment increases, then, in the short period, the reward per unit labour in terms of wage goods must, in general, decline and profit increase.'

Keynes recommendation for increasing employment would be identical to theirs, that is, to decrease wages and increase profit. If profit-maximizing behavior motivated capitalists to hire laborers to the point where their wage equaled the value of their marginal product then lowering the real wage of workers was the only answer to unemployment.

Keynes argued in a very unconvincing way that the real wages could be decrease in one of the two ways.

1. Money wage rate could be decreased while the wage goods prices remained constant.
2. The prices of wage goods could be increased while the money wage rate remained constant.

The second method is more acceptable to the workers according to Keynes. The argument is unconvincing because in their wage negotiations, workers are generally as concerned with the cost of living as they are with their money wages. When workers are in a strong bargaining position, they generally insist that money wage increases compensate them for any increase in the cost of living. Moreover, in their

generally destitute condition in the 1930s, millions of workers aggressively searched for any job that would pay a livable wage.

Before the great depression, if the workers were fully employed and get their wages according to the value of marginal productivity then in 1930s there were sharp decline in employment and marginal productivity of labor increased according to the logic of marginal productivity theory of distribution, which Keynes fully accepted. But real wages did not increased in 1930 and employed workers were receiving real wages below their marginal productivity.

Efficacy of Keynes policies

1930 is the year of great depression. In 1936, Keynes 'General Theory' was published.

During World War 2, government began to increase production rapidly and unemployment began to melt away. And most capitalist economies were rapidly transformed from situations of severe unemployment to shortages of labor.

Between 1939 and 1944 the output of the manufacturing, mining and construction industries doubled and productive capacity increased by 50 percent. The American economy produced 296,000 planes, 5,400 cargo ships, 6,500 naval vessels, 64,500 landing craft, 86,000 tanks, 2,500,000 trucks, and massive amounts of other supplies and materials of warfare.

After 1945, the majority of politicians joined economists in proclaiming the new Keynesian orthodoxy. In 1946, Congress passed the Employment Act, which legally obligated the government to use its taxing, borrowing, and spending powers to maintain full employment.

Most economists believed that this wartime experience proved Keynes's ideas. The massive government expenditures of the war pulled the American economy out of a decade of stagnation and depression.

In 1948-49, a "recession" lasted for eleven months; in 1953-54, for thirteen months; in 1957-58 for nine months; and in 1960-61 for nine months. While the Vietnam War gave added stimulus to the American economy in the 1960s, by the end of that decade the old pattern had set in again. The recession of 1969-71 lasted about two full years. There were few more recession in American history.

Still, since World War II, unemployment rates have not approached those of the Great Depression (which reached 20-25 percent by official figures and would

probably have been closer to 30-35 percent had more realistic methods of defining unemployment and gathering statistics been used).

- Unemployment in the 1950s and early 1960s averaged 4.6 percent.
- Vietnam War, in the mid-1960s the unemployment rate fell to 3.5 percent.
- Through the 1970s and 1980s, it ranged between 5 percent and 10 percent.
- In the 1990s, the rate peaked at 7.9 percent just after the 1990-91 recession and steadily declined to 3.9 percent by the close of the decade.
- The rate peaked at 6.3 percent by 2003 in the recession of the early 2000s.
- In the latest recession, beginning at the end of 2007, the unemployment rate in the United States went over 10 percent.

Although the post-World War II record is certainly less impressive than the optimistic vision of many Keynesian economists writing in the 1940s, given the stagnation and near disintegration of capitalism in the 1930s, it may be said that for over seven decades Keynesian policies have worked tolerably well.

The Military Economy:

If one analyzes the "Multiplier Effect" of aggregate demand created by military expenditures, the impact was much larger. In 1947, military expenditures accounted, directly and indirectly for 24.4% of aggregate demand in 1947 and 22.2% in 1971. Between 1947 and the mid-1970s, the American government spent nearly \$2 trillion on military expenditures. In the 1970s military spending grew at a lower rate but it exploded in the 1980s.

The recession of 1981-82 was the worst since the 1930s. Between 1980 and 1987, American military expenditures more than doubled. The U.S. Defense Department increased its employment by 255,000 new jobs between 1981 and 1985. The prosperity of the 1980s was a huge growth in militarism. President Bush's budget proposal for the fiscal year 1991 included over \$303 billion directly for defense and another \$65 billion for international affairs, science, space, and technology, and veterans' benefits, most of which are disguised military spending. Defense spending remained high from 1990 to 1999. The correlation between changes in military spending and the cyclical instability of capitalism is very significant.

Two facts clearly emerge from the study of American militarism:

1. Militarism offers several unique and distinct advantages to capitalists over nearly any other form of government spending.
2. Militarism has become so thoroughly enmeshed in the structure of the American economy that it is increasingly inconceivable that it can ever be extirpated short of a complete restructuring of the entire American social, economic, and political system.

Militarism offers innumerable advantages to large corporations:

- It stimulates aggregate demand without redistributing income from the rich to the poor.
- There can never be too much elaborate weaponry; government-financed research constantly renders military hardware obsolete, and horror stories convince most of the public that continued escalation of the arms race is absolutely necessary for survival.
- The capital goods industry-the most volatile and unstable segment of a capitalist economy is kept operating at near capacity by military production, and, yet, this does not increase the productive capacity of the capitalist economy. As a result, militarism does not worsen the persistent problem of excess aggregate productive capacity.
- All military production is done by giant private corporations or subcontracted to smaller private corporations. It does not compete with private profit making. On the contrary, it reduces the anarchy of the free market by giving corporations a stable core of demand not subject to the vagaries of the market.
- Although precise statistics to substantiate this are unobtainable, most evidence points to military production being more profitable than production for the free market.
- As Rosa Luxemburg pointed out, large military establishments are necessary to maintain and enlarge the worldwide "Sphere of influence" of capitalist countries, in which profitable foreign investment and very favorable terms of trade can be assured.
- As Veblen pointed out, jingoism, patriotism, and militarism are perhaps the most effective means of keeping workers docile and promoting the view

among workers that their interests are in harmony with the interests of the capitalists.

A large reduction in the output of military hardware or in the number or size of these military bases can economically ruin whole communities. On the other hand, the slightest hint of a reduction in militarism brings powerful protests from military leaders, large corporations, politicians, and labor union leaders. Many of the largest and most powerful of American corporations are heavily dependent on military contracts.

The Debt Economy:

Keynes had demonstrated how demand, rather than resources, could constrain an economy's output. A capitalist economy could operate below its potential for a significant period of time. Businesses would only increase output, and their utilization of existing resources if there was a high probability of making sales which translated into profits.

Keynes identified how investment could fall short of the amount of saving possible at the economy's potential level of income, thereby placing a demand constraint on the economy. If the rate of interest could not fall low enough to rectify this situation then other sources of demand would need to be found to fill the gap. This is the basis for Keynesian fiscal policy. Expansion of government expenditures financed by borrowing would be more effective than financing by taxation.

The United States government came out of World War II with an enormous amount of debt. In 1946, the federal government's debt relative to GDP stood at 109%. The nominal value of the federal debt continued to increase, it fell rather consistently relative to GDP until the mid -1970s. From 1946 to 1974, the U.S. experienced an average annual GDP growth of around 7%. The increased government expenditures due to World War II accompanied by the recovery from the Great Depression was supporting evidence for Keynesian fiscal policy. Federal debt relative to GDP continued to increase until 1993. The annual growth rate of the federal debt fell consistently after 1992 and turned negative by 1998. Just as the growth in federal debt began to slow, there occurred a significant increase in the annual growth rate of business debt from negative 2% in 1992 to positive 11% in 1998. U.S. households had come out of World War II with a low debt load which increased steadily until the mid-1960s before flattening out until the mid-1980s.

Keynes emphasized the importance of demand in determining output and employment, he was actually quite conservative fiscally. In terms of the growth of private debt, Keynes again appeared somewhat shortsighted. Household debt in the United States began to increase dramatically in 2000, averaging 10% annual growth until 2007. In 2009, the growth in household debt fell to zero, as credit markets froze, turning to negative growth when the crisis hit. The policy response to the crisis followed Keynesian lines as central banks pumped money into their economies and lowered interest rates. Beyond these measures, the lack of growth in private debt was replaced with growth in public debt.

Ideological Foundations of Keynes's Ideas:

Keynes wanted to drop the assumption of the automaticity of the market in order to save capitalism from self-destruction. He wanted to maintain faith in the marginal productivity theory of distribution and faith in the allocative efficiency of the market. He wanted government to intervene as little as possible into capitalists' quest for profits, and then only to avert disaster. However, he did mention as an aside that he personally preferred a less extreme degree of inequality in the distribution of wealth and income. Keynes believed that there was social and psychological justification for significant inequalities of incomes and wealth. He had faith, however, that there were forces at work in a capitalist system that automatically tended, in the long run, to mitigate the extremes of inequality. He believed that in the short run, in order to stimulate employment, real wages would have to fall and profits would have to increase.

Both Ricardo and Keynes were much influential in their respective sectors and their ideas opened new stages of economic theories that made possible to analyze the critical issues of today's world. The theory of Comparative Advantage made possible to create international trade among different countries. On the other hand, Keynes's theory of government intervention in economy has ensured how the economy would function and regulated as well as monitored. It is a crying need to practice their ideas and develop further theories to make better understanding of future social and economic problems and solutions.